

Energy and greenhouse gas impacts of mining and mineral processing operations

T. Norgate*, N. Haque

CSIRO Minerals Down Under Flagship, Box 312, Clayton South, Victoria 3169, Australia

ARTICLE INFO

Article history:

Received 16 June 2009

Received in revised form

23 September 2009

Accepted 24 September 2009

Available online 2 October 2009

Keywords:

Life cycle assessment

Mining and mineral processing

Embodied energy

Greenhouse gases

ABSTRACT

Life cycle assessments of the mining and mineral processing of iron ore, bauxite and copper concentrate were carried out, focussing on embodied energy and greenhouse gas emissions. The results showed that loading and hauling make the largest contributions to the total greenhouse gas emissions for the mining and processing of iron ore and bauxite. In the case of copper ore, the crushing and grinding steps make the largest contribution to the total greenhouse gas emissions for the production of copper concentrate. These results indicate that efforts to reduce the increased greenhouse gas emissions from mining and mineral processing, anticipated in the future as a result of falling ore grades and more finer-grained deposits, should focus on loading and hauling for iron ore and bauxite, while for copper ore the focus should be on grinding. There are a number of new and emerging technologies that could be expected to assist in this task, and these include high pressure grinding rolls and stirred mills for grinding, and further advances in diesel engine technology for loading and hauling applications.

Crown Copyright © 2009 Published by Elsevier Ltd. All rights reserved.

1. Introduction

The ever increasing global demand for consumer goods means that the production of primary metals can be expected to increase well into the future, despite society's best efforts in recycling and dematerialisation (broadly defined as the reduction in the amount of energy and materials required to serve economic functions, e.g. production of consumer goods or the provision of services). Thus well into the future, metals will be supplied from a combination of primary metal produced from newly mined ores and recycled metals, though the amount of metal recycled will continue to increase. The mining, mineral processing and metal production sector, like other industrial sectors, is coming under increased pressure to reduce the amounts of energy it consumes and greenhouse gases it emits. This has led to the application of life cycle assessment methodology (discussed later) to the production life cycles of most metals [1]. However, most life cycle assessments of metal production processes do not consider the mining and mineral processing stages in any detail, largely due to a lack of publicly available data. This generally means that mining and mineral processing are lumped together as a single stage in the metal life cycle. This approach does not usually introduce any significant errors due to the relatively small contribution that the mining and mineral processing stages make to the 'cradle-to-gate' environmental impacts of many metal production processes,

particularly with regard to impacts such as embodied energy and greenhouse gas emissions. This is illustrated in Fig. 1, which compares the embodied energy of the mining and mineral processing stages to the downstream metal extraction (smelting and refining) stages for iron, aluminium and copper [1].

However, it has been pointed out previously [2,3] that the grades (i.e. metal content) of metallic ores have been falling globally for some time, and that this will have a significant effect on the amount of energy required for mining and processing of these lower grade ores due to the additional amount of material that must be treated in these stages. On the other hand, lower ore grades will not significantly increase the energy consumption of the downstream metal extraction and refining stages of many metals (e.g. copper), as a concentrate of fixed grade is produced for downstream processing, irrespective of the initial ore grade. This is illustrated in Fig. 2 for copper produced pyrometallurgically¹. In addition to falling ore grades, many of the newly discovered ore deposits are complex and finer-grained, requiring grinding to finer sizes to liberate the valuable or waste minerals in order to achieve separation and concentration. This will also increase the energy consumption of the mineral processing stage, and the combined effects of declining ore grades and finer grind sizes on the embodied energy of copper metal produced pyrometallurgically are shown in Fig. 3.

¹ Pyrometallurgical processing involves smelting at high temperature, while hydrometallurgical processing involves leaching generally at ambient temperatures.

* Corresponding author. Tel.: +61 03 9545 8574; fax: +61 03 9562 8919.
E-mail address: terry.norgate@csiro.au (T. Norgate).

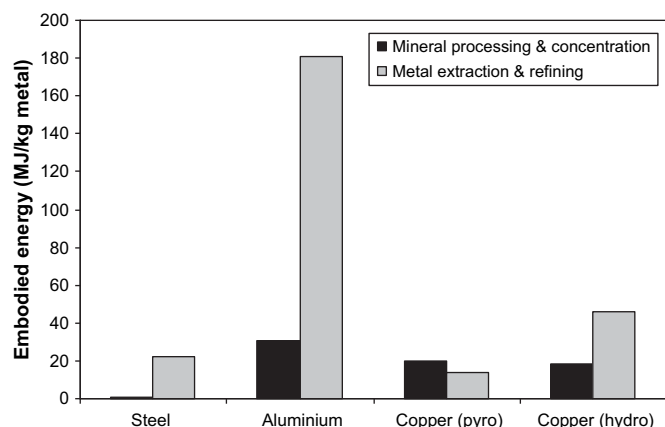


Fig. 1. Processing stage contributions to embodied energy of steel, aluminium and copper production.

Thus it can be expected that the environmental impacts, particularly energy consumption and greenhouse gas emissions, of mining and mineral processing for many metals will become much more significant in the future than they currently are. It is therefore important that the contributions of the various processing steps that make up these stages be quantified, with the major contributing steps being identified in order that efforts to reduce these environmental impacts be focussed on these steps. With this objective in mind, life cycle assessments (LCAs) of the mining and mineral processing of iron, aluminium (bauxite) and copper ores were carried out. These three ores were chosen for this study mainly due to the significant amount of these ores mined annually in Australia, as shown in Table 1. In the case of iron ore and bauxite, these ores are largely transported to downstream metal extraction plants without any significant beneficiation because of their relatively high ore grades as mined (e.g. typically 60% for iron ore and 22% for bauxite in Australia). However, copper ore generally undergoes beneficiation (concentration) to produce a concentrate (typically in the order of 30% Cu) for downstream metal extraction as pointed out above. This paper presents the results of these LCAs, with breakdown of the results to show the contributions of the various processing stages to the overall impact. Some possible technologies to reduce the energy and greenhouse gas impacts of mining and mineral processing operations are also described in the paper.

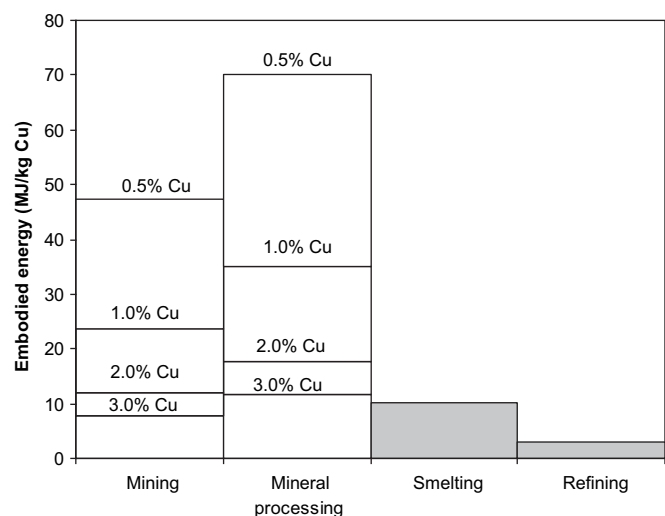


Fig. 2. Effect of ore grade on embodied energy for pyrometallurgical copper production.

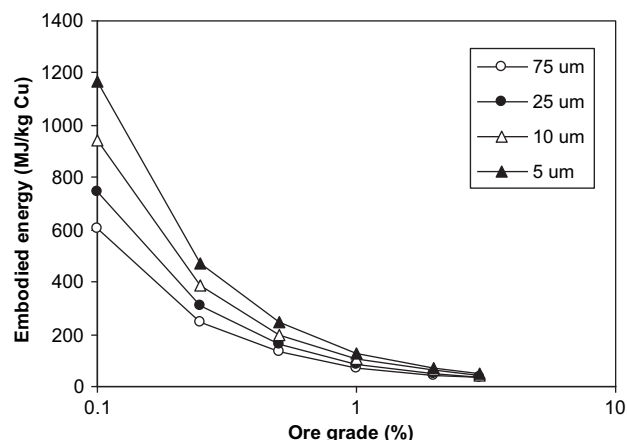


Fig. 3. Combined effect of ore grade and grind size on embodied energy for pyrometallurgical copper production.

2. Mining and mineral processing operations

The extraction of metallic ores involves both surface (open-pit) and underground mining techniques. The method selected depends on a variety of factors, including the nature and location of the deposit, and the size, depth and grade of the deposit. Underground mining requires more energy than surface mining due to greater requirements for hauling, ventilation, water pumping and other operations. Surface mining accounts for the majority of mining, although most of the copper ore produced in Australia comes from underground mines. The various mining and mineral processing stages are described in detail below.

2.1. Drilling

Drilling is the act or process of making a cylindrical hole with a tool for the purpose of exploration, blasting preparation, or tunneling. Drilling equipment includes explosive loader trucks, diamond drills, rotary drills, percussion drills and drill boom jumbos. Drills are run from electricity, diesel power and to a lesser extent, indirectly from compressed air. The energy is used to power components of the drill that perform tasks such as hammering and rotation. The number of drilling machines is about 2–6 depending on the mine production capacity.

2.2. Blasting

Blasting uses explosives to aid in the extraction or removal of mined material by fracturing rock and ore by the energy released during the blast. The energy consumed in the blasting process is derived from the chemical energy contained in the blasting agents. This sets blasting apart from other processes, which are powered by traditional energy sources, such as electricity and diesel fuel. A common explosive used for mining is ammonium nitrate/fuel oil (ANFO) mixture. The powder factor is the amount of explosives used per unit of rock blasted, and varies depending on the rock type and strength. The blast holes are detonated with a nonel (non-electric)

Table 1
Annual tonnages of ores mined in Australia.

Ore mined	Mtpa ore	References
Iron ore	236	Australian mineral statistics (abare.gov.au)
Bauxite	57	Australian Mineral Statistics (abare.gov.au)
Copper ore	50 ^a	Australian Mineral Statistics (abare.gov.au)

^a Based on 896 ktpa contained Cu and average Australian copper ore grade of 1.8% (calculated by authors).

Table 2
Inventory data for mining and mineral processing operations used in study.

Ore mined	Mining method	Stage	Inventory		
			Item	Value	Units
Iron ore	Open-cut	Drilling	Diesel	0.03	kg/t ore
		Blasting	Explosives ^a	0.5	kg/t ore
		Loading & hauling	Diesel	2.2	kg/t ore
		Crushing & screening	Electricity	2.5	kWh/t ore
		Stacking & reclaiming	Electricity	0.5	kWh/t ore
		Rail transport	Diesel	0.5	kg/t ore
		Port operations ^b	Electricity	0.8	kWh/t ore
		Overall	Water	0.21	m ³ /t ore
			Diesel ^c	3.4	kg/t ore
				135	MJ/t ore
			Electricity	3.8	kWh/t ore
			Explosives	0.5	kg/t ore
			Waste rock	1.3	t/t ore
Bauxite	Open-cut	Drilling	Diesel ^d	0.03	kg/t bauxite
		Blasting	Explosives ^a	0.3	kg/t bauxite
		Loading & hauling	Diesel	0.9	kg/t bauxite
			Electricity	0.1	kWh/t bauxite
		Crushing & blending	Electricity	1.7	kWh/t bauxite
		Beneficiation	Electricity	0.1	kWh/t bauxite
		Overall	Water	0.30	m ³ /t bauxite
			Diesel	0.93	kg/t bauxite
				38	MJ/t bauxite
			Electricity ^e	2.0	kWh/t bauxite
			Explosives	0.3	kg/t bauxite
			Waste rock	0.3	t/t bauxite
Copper ore	Underground	Drilling	Diesel	0.7	kg/t ore
		Blasting	Diesel	0.1	kg/t ore
			Explosives ^a	0.4	kg/t ore
		Loading & hauling	Diesel	2.0	kg/t ore
		Ventilation	Electricity	8	kWh/t ore
		Dewatering	Electricity	3.8	kWh/t ore
		Crushing & grinding	Electricity	18.5	kWh/t ore
		Concentrating	Copper ore ^f	16.2	t ore/t concentrate
			Electricity	7.5	kWh/t ore
			Reagents ^g	1.7	kg/t ore
			Grinding media	1.4	kg/t ore
			Tailings	37	t/t concentrate
		Overall	Copper ore	16.2	t ore/t concentrate
			Water	0.51	m ³ /t ore
			Diesel	2.8	kg/t ore
				115	MJ/t ore
			Electricity ^h	46.4	kWh/t ore
			Explosives	0.4	kg/t ore
			Reagents	1.7	kg/t ore
			Grinding media	1.4	kg/t ore
			Waste rock	0.03	t/t ore
			Tailings	2.3	t/t ore

^a Ammonium nitrate/fuel oil (ANFO) – 94% AN & 6% FO. Production of AN = 7.1 MJ/kg, 1.2 kg CO₂e/kg; ANFO = 6.6 MJ/kg, 1.4 kg CO₂e/kg.

^b Includes train unloading, stacking, reclaiming & shiploading.

^c Includes miscellaneous equipment 0.7 kg/t ore.

^d Assumed same as for iron ore.

^e Includes miscellaneous/auxiliary equipment 0.1 kWh/t bauxite.

^f Based on ore grade 1.8% copper, concentrate grade 27.3% copper, 93.7% copper recovery.

^g Dependent on ore mineralogy - assumed to comprise 80% lime, 12% xanthate, 8% sodium cyanide.

^h Includes miscellaneous/auxiliary equipment for underground mining 8.6 kWh/t ore.

device for firing. Blasting frees ore from the host rock and reduces the size of ore before it undergoes crushing and grinding, thereby reducing the energy consumption of crushing and grinding processes. Optimising blasting techniques will therefore produce downstream energy savings.

2.3. Ventilation

Ventilation is the process of bringing fresh air to the underground mine workings while removing stale and/or contaminated air from the mine and also for cooling work areas in deep underground mines. The mining industry uses fan systems for this purpose.

2.4. Dewatering

Dewatering is the process of pumping water from the mine workings. Pumping systems are large energy consumers. This study assumes that centrifugal pumps are used for dewatering the mine during ore extraction.

2.5. Loading and haulage

In open-pit mines, the broken rocks are generally excavated by either front-end loaders, excavators or shovels and loaded into a dump truck for haulage to the processing plant. Most mines have a

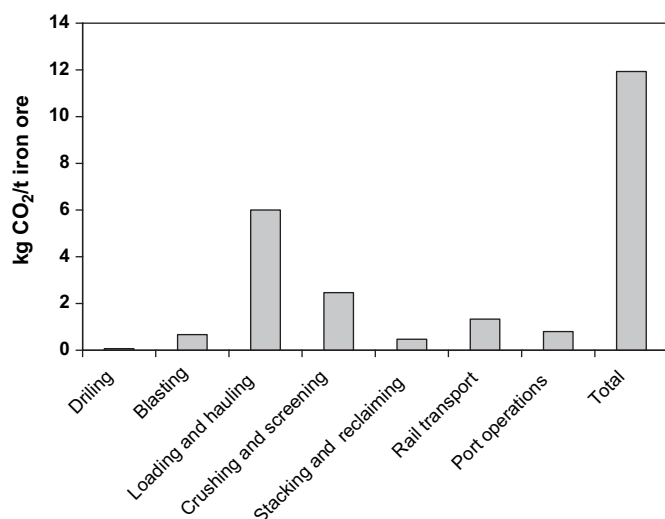
Table 3

Energy and greenhouse gas results from study.

Metal ore	Energy (MJ/t ore ^a or conc. ^b)	GWP (kg CO ₂ e/t ore ^a or conc. ^b)
<i>Iron ore</i>	(%)	(%)
- Drilling	1.3 (0.9)	0.1 (0.8)
- Blasting	3.3 (2.2)	0.7 (5.9)
- Loading & hauling	92.1 (60.3)	6.0 (50.5)
- Crushing & screening	23.1 (15.1)	2.5 (21.0)
- Stacking & reclaiming	4.6 (3.0)	0.5 (4.2)
- Rail transport	20.9 (13.7)	1.3 (10.9)
- Port operations	7.4 (4.8)	0.8 (6.7)
- Total	152.7	11.9
<i>Bauxite</i>	(%)	(%)
- Drilling	1.2 (2.2)	0.1 (2.0)
- Blasting	2.0 (3.6)	0.4 (8.2)
- Loading & hauling	36.1 (65.8)	2.6 (53.1)
- Crushing & screening	14.7 (26.8)	1.7 (34.7)
- Beneficiation	0.9 (1.6)	0.1 (2.0)
- Total	54.9	4.9
<i>Copper concentrate</i>	(%)	(%)
- Drilling	720 (8.6)	30.8 (4.9)
- Blasting	43 (0.5)	9.1 (1.4)
- Loading & hauling	2059 (24.7)	88.1 (14.0)
- Ventilation	1417 (17.0)	127.0 (20.2)
- Dewatering	673 (8.1)	60.3 (9.6)
- Crushing & grinding	3277 (39.4)	293.7 (46.8)
- Concentrating	140 (1.7)	19.2 (3.1)
- Total	8329	628.2

^a Iron ore and bauxite.^b Copper concentrate.

loading fleet including wheel loaders, shovel units and excavators. The wheel loaders have a capacity ranging from 50 to 90 tonnes, while the shovel units and excavators have capacities ranging from 200 to 250 tonnes. The haulage units typically include off-road dump trucks with carrying capacities ranging from 150 to 300 tonnes of rocks. Typical number of these haul trucks can be from 10 to 22 depending the mine size. Typical energy consumption data for the loading and hauling fleet were collected as part of this study. Much of the equipment used in the transfer or haulage of materials in mining is powered by diesel engines. Diesel technologies are highly energy intensive, accounting for 87% of the total energy consumed in materials handling [4].

**Fig. 4.** Stage contributions to GWP for iron ore production.

2.6. Auxiliary equipment

On most mine sites, there is other equipment such as dozers, graders, excavators and water tankers. They are used for road construction, maintenance and dust suppression within the mine site. It is assumed that these units use diesel fuel for their operation.

2.7. Crushing and grinding

Crushing is the process of reducing the size of run-of-mine material into coarse particles (typically coarser than 5 mm). The efficiency of crushing in mining depends on a number of factors including the efficiency of upstream processes (rock fragmentation due to blasting or digging in the extraction process) that in turn, has a significant effect on downstream processes (grinding or separations). Grinding is the process of reducing the size of material into fine particles (often below 0.1 mm or 100 µm). As with crushing, the efficiency of grinding is influenced by upstream processes that fragment the rock prior to the grinding stage. In the case of both crushing and grinding, estimates of their energy efficiency in the literature vary widely, with energy efficiencies as low as 1% being reported for grinding [5]. Crushing and grinding plants are usually powered by electric motors, with the electricity often generated onsite using a diesel fuel-based engine and generator. Crushing plants can include primary, secondary and tertiary crushers, while grinding plants can include SAG², rod and ball mills.

2.8. Separations

The separation of mined material is achieved primarily by physical separations rather than chemical separations, where valuable substances are separated from undesired substances based on the physical properties of the materials. There is a wide variety of equipment used for separation processes, the largest energy-consuming separation method amongst these being centrifugal separation for coal mining, and flotation for metals and minerals mining. Flotation machines are designed to isolate valuable ore from other non-valuable substances. The surfaces of mineral particles are treated with chemical reagents to make some selectively hydrophobic. The ore is suspended in water that is mechanically agitated and aerated. The mineral particles that have been rendered hydrophobic attach to air bubbles and rise to the surface where they can be collected. In the case of iron ore mining, screening is the most common separation process which is used to separate the ore into lump and fines streams, while magnetic separation is used to separate magnetite from gangue.

2.9. Environmental management

Mining and mineral processing activities are an integral part of complex material cycles in society, which in turn, interact with natural material cycles and ecosystems. Hilson [6] has listed 27 potential environmental impacts of such interactions. The sustainability of the minerals industry is about managing these cycles in ways that maximise the value to society while minimising negative impacts, be they economic, social or environmental. Thus the mining industry has increasingly embraced the concepts of “environmental management” [7–9], “sustainable development” [10–12] and “corporate social responsibility” [13] over the last two decades. Environmental management has evolved over the years through a series of successive paradigms: (a) passive environmental management; (b) reactive environmental management or

² Semi-autogeneous grinding mills.

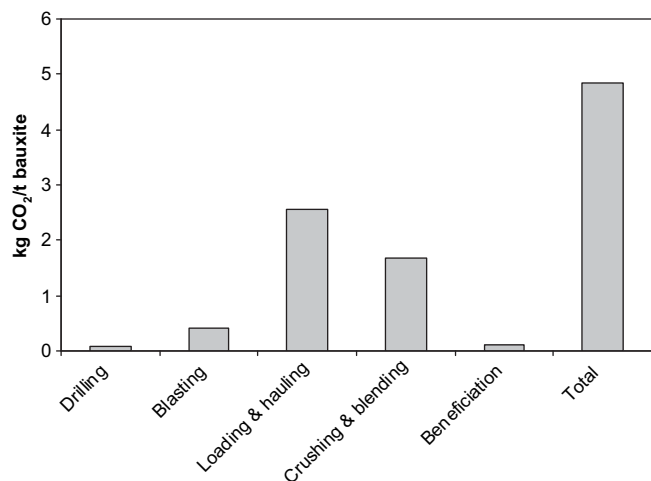


Fig. 5. Stage contributions to GWP for bauxite production.

end-of-pipe approaches; (c) proactive environmental management or cleaner production [6]. Cleaner production explicitly targets the reduction of environmental impacts along a product's life cycle [14]. Hilson [6] suggests that practices to implement the concept of cleaner production in mining can be divided into three separate categories: (a) managerial changes – environmental management-related initiatives that improve the overall efficiency of operations, e.g. implementation of an environmental management system (EMS); (b) policy changes – corporate environmental policies, environmental audits; (c) physical changes – technological modifications, implementation of state-of-the-art equipment. Several mining companies have begun integrating LCA into their EMS and this methodology is described in the next section.

The Australian mining industry is regarded as a global leader in the area of mining EMS design and implementation. The Minerals Council of Australia – the national association of the Australian minerals industry – developed the Australian Minerals Industry Code for Environmental Management, which marked a significant step toward addressing environmental performance and public accountability at minerals operations. The Code was initially launched in December 1996, and underwent a substantial review process in 1999 which resulted in a revised Code being released in January 2002. This Code was formally retired in January 2005 and replaced by “Enduring Value”, which is the Australian Minerals

Industry Framework for Sustainable Development. As at December 2007, 34 companies were signatories to “Enduring Values”. While there are a number of environmental issues that mining and mineral processing operations will have to increasingly contend with, the three major ones are likely to be energy and greenhouse gas emissions and water consumption (particularly in Australia) – energy and greenhouse gas emissions due to concerns over climate change impacts, and water due to strong competing demands for water resources [15]. The energy and greenhouse gas issues are the main focus of the LCA described below.

3. Life cycle assessment

Life Cycle Assessment (LCA) is a relatively new methodology that can be used to assess the environmental impact of various activities, products and processes objectively. LCA covers the consecutive and interlinked stages of a product or process system, from raw material acquisition or generation from natural resources through to final disposal. It essentially involves the compilation of an inventory of relevant environmental exchanges during the life cycle of a product and evaluating the potential environmental impacts associated with those exchanges. The full product life cycle is usually divided into the following stages:

- cradle to entry gate (raw material extraction to refining);
- entry gate to exit gate (product manufacture);
- exit gate to grave (product use, recycling and disposal).

Based on impact assessment, two types of LCA can be distinguished, problem-oriented (mid points) or damage-oriented (end points) [16]. The LCAs of mining and mineral processing operations described in this paper were all based on the problem-oriented approach and were carried out in accordance with the international standard LCA guidelines [17] using the SimaPro software package.

3.1. Previous mining LCA studies

There have been a number of studies published regarding the application of LCA methodology to mining and mineral processing. Yellishetty et al. [18] carried out a critical review of existing LCA methods in the mining and metals sector and discussed a number of related issues. Suppen et al. [9] provided a general overview of the Mexican mining industry and described strategies being implemented to incorporate sustainable development principles, including the development of a national base metals life cycle inventory. Durucan et al. [19] developed a mining life cycle model incorporating an inventory database that enables mining LCA studies to be carried out and described its application to an open-pit bauxite mine in Hungary. Adachi and Mogi [20] described the development of a mining life cycle inventory database focussing mainly on greenhouse gas emissions and its application to copper and zinc metal production, while Mangena and Brent [21] presented the results of a ‘cradle-to-gate’ LCA study of coal production from four mine sites in South Africa. Awuah-Offei et al. [22] carried out an LCA study of belt conveyor and truck haulage systems in an open-pit hard rock gold mine and showed that the greenhouse gas emissions of the belt conveyor system was appreciably greater than that of the truck haulage system for the same functional unit (hauling 4000 t/h of rock/ore), although the acid rain gas emissions were less for the belt conveyor system. However, while some of the above publications include results from actual LCA case studies, none of them give a breakdown of the results to show the contributions of the various mining and mineral processing steps, which was the objective of the present study.

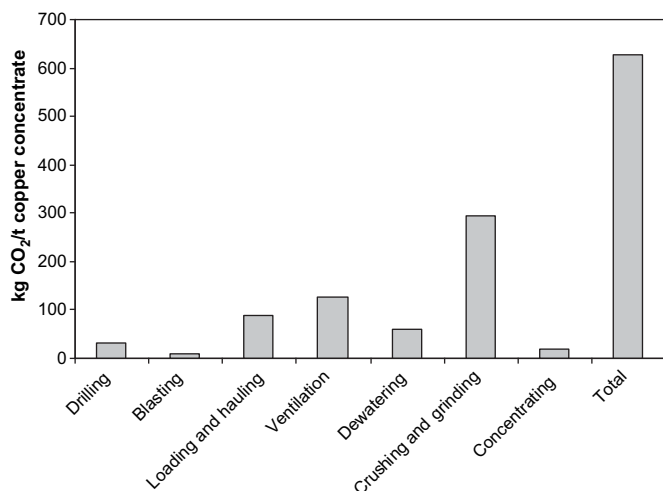


Fig. 6. Stage contributions to GWP for copper concentrate production.

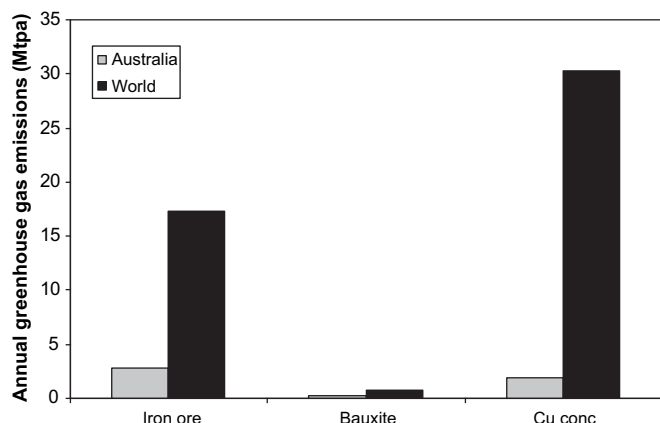


Fig. 7. Annual greenhouse gas emissions from iron ore, bauxite and copper concentrate production.

3.2. This study

3.2.1. Goal and scope

The goal of this study was to determine the life cycle-based energy requirement and associated greenhouse gas emissions of selected mining and mineral processing operations to assist the Australian minerals industry in identifying potential areas for improvement of their environmental performance, particularly as ore grades fall in the future. The study covers the upstream part of the cradle-to-gate life cycle of the mining and processing of the ores of iron, aluminium (bauxite) and copper. In the case of iron ore, transport of lump and fines to the shipping port was also included, as these mines are usually located some distance inland in Australia. The iron ore and bauxite mines were open-pit (i.e. surface) mines while the copper ore mine was underground. Mine site rehabilitation was not included in the study. The functional unit assumed for the study was 1 tonne of ore or concentrate ready for ship loading. The impact categories considered were gross energy requirement (or embodied energy) and greenhouse gas emissions (expressed as global warming potential, GWP). The Australian impact method with normalisation (Version 1) has been used for calculation of the results based on Australian data.

3.2.2. Inventory data

There is little published data in this area. The LCA inventory data used in the study were derived from a number of published sources for bauxite [23,24] and copper ore [4,25,26], supplemented by other mining inventory data collected by the authors. In the case of iron ore, calculations were made by the authors and these are described below. All inventory data used in the study are tabulated in Table 2. There was no need for any co-production allocation, since all mine sites considered produced only one product (i.e. iron ore, bauxite or copper concentrate). Given the limited amount of publicly available data in this area, this study is considered to be a preliminary, or first-approach, investigation to assess the relative contributions of the various stages to the energy and greenhouse gas footprints of the selected mining and mineral processing operations. It is anticipated that the data will be refined further over time, with the LCA results being progressively updated as industry stakeholders release in-house inventory data.

3.2.3. Iron ore calculations

Four Australian hematite/goethite mine sites (i.e. Brockman, Paraburdoo, Hopes Down, Yandicoogina) were chosen for this analysis to derive representative information for hematite/goethite iron ore mining in Australia. These mine sites were chosen as they range from small to large mine sites in terms of production capacity. Data (i.e.

number of machines, annual production, etc) for each mine site were collected from company websites and a published survey of each mine site [27]. From the respective mining and mineral processing flowsheets, each processing unit was identified. Material and equipment required for each unit were then derived from published data. The number of the various equipment items, their capacities and energy consumption were collected for each mine site from several sources, including mining and equipment manufacturing company websites and published survey [27]. Data for each mine site were weighted according to production tonnage to give the mean values shown in Table 2. As an additional cross-check, the calculated results were compared with published values where available, e.g. [28].

4. Results

The energy and greenhouse gas results from the LCA study of the various mining and mineral processing operations are given in Table 3 and shown graphically in Figs. 4–6. The greenhouse gas emissions were 11.9 and 4.9 kg CO₂e/t ore for iron ore and bauxite respectively, while for copper concentrate they were 628 kg CO₂e/t concentrate. Based on the inventory data in Table 2, the latter figure corresponds to 38.8 kg CO₂e/t ore, which is not too different to the value of 32 kg CO₂e/t ore reported for base metal ores [29]. The embodied energy values were 153 MJ/t ore, 55 MJ/t ore and 8329 MJ/t concentrate for iron ore, bauxite and copper concentrate respectively. These results show that loading and hauling made the largest contributions (approximately 50%) to the total greenhouse gas emissions for the mining and processing of iron ore and bauxite. In the case of copper ore, the lower ore grade compared to iron ore and bauxite means that additional processing (i.e. concentration) is required to produce a concentrate of suitable grade, and in this case it is the crushing and grinding (particularly the latter) steps for this additional concentration that make the largest contribution (approximately 47%) to the total greenhouse gas emissions for the production of copper concentrate. Explosives made only a small (1–8%) contribution to the overall greenhouse gas emissions, amounting to 0.4 and 0.7 kg CO₂e/t for bauxite and iron ore respectively, and 9.1 kg CO₂e/t concentrate (or 0.6 kg CO₂e/t ore) for copper concentrate.

The annual greenhouse gas emissions resulting from the mining and processing of these ores in Australia can be estimated in broad terms by combining the results in Table 3 with the annual production data given in Table 1, and are shown in Fig. 7. The production of copper concentrate was responsible for 1.9 Mtpa of CO₂e, while the production of iron ore and bauxite was responsible for 2.8 Mtpa and 0.3 Mtpa of CO₂e respectively. These calculations can be extended to the global scene by using world annual production tonnages of these materials. These tonnages were estimated by the authors from the reported world annual production of these metals (steel, aluminium and copper) together with some simplifying assumptions³, and further assuming that previously reported [32–34] LCA inventory data for these metals applies to all of these production volumes. The estimated world annual production of iron ore, bauxite and copper concentrate were 1460 Mtpa, 170 Mtpa and 48 Mtpa respectively. Combining these values with the results in Table 3 gives the global results shown in Fig. 7, with the production of copper concentrate being responsible for 30 Mtpa of CO₂e, while the production of iron ore and bauxite was responsible for 17 Mtpa and 0.8 Mtpa of CO₂e respectively. Thus iron ore is the

³ 70% of world steel production is via the integrated route from iron ore [30], and 80% of world copper production is from pyrometallurgical processing of copper concentrate [31].

Table 4
Potential energy savings for mining and mineral processing [4].

Stage	Current energy consumption ^a (MJ/t ore)	Practical minimum energy consumption ^a (MJ/t ore)	Potential energy savings (MJ/t ore)	(%)
Drilling	2.7	1.5	1.2	43
Blasting	11.5	4.8	6.7	59
Loading & hauling ^b	87.2	41.5	45.7	52
Ventilation	17.5	13.6	3.9	23
Crushing	6.9	3.6	3.3	48
Crushing & grinding	220.6	63.2	157.4	71

^a Primary energy (i.e. all electrical energy converted to thermal energy).

^b Diesel equipment.

more significant greenhouse gas contributor in Australia, while copper concentrate is more significant on a global scale. However, given the uncertainty in the calculations, these results should be considered as very broad comparisons only.

5. Technologies to reduce energy and greenhouse gas impacts of mining

It has been reported [4] that the metal mining industry in the United States has the potential to reduce energy consumption by about 61% from current practice to the best-estimated practical minimum energy consumption. This reduction was made up of a 21% reduction by implementing best practices and a 40% reduction from research and development that improves energy efficiency of mining and mineral processing technologies. Reported [4] current and practical minimum energy consumptions, including potential energy savings for the various stages of the mining and mineral processing life cycle are given in Table 4. In some cases, the practical minimum energy consumption was determined from published estimates of future attainable equipment efficiencies. In other cases where no published practical minimum target could be found, it was assumed that the practical minimum energy is two-thirds of the way between best practical energy requirement and the theoretical energy requirement. Calculations for diesel engines, motors and pumps where published data were available, showed that the two-thirds approximation provided good estimates of practical minimum energy consumptions compared to the published data [4]. It can be seen from Table 4 that crushing and grinding offers the greatest opportunity for energy savings per tonne of ore processed, followed by loading and hauling. Assuming the energy savings given in Table 4 will be realised in the future, their effects on the greenhouse and embodied energy impacts of mining were estimated by incorporating these energy savings into the respective LCAs⁴, and the results are shown in Figs. 8 and 9. As a conservative approach, no energy savings were assumed for those processing stages listed in Table 2 but not included in Table 4.

Some of the new and emerging technologies that could contribute to these energy savings include high pressure grinding rolls [35] and stirred mills [36] for grinding, as well as more advanced blasting techniques [37] and further advances in diesel engine technology for loading and hauling applications [4]. Another possible approach to improving the energy efficiency of loading and hauling is by optimising pit and mine design to reduce haulage requirements. For example, in-pit mobile crushing and conveying systems for open-pit mining reportedly [38,39] eliminates the need for trucks by having the shovel feed the run-of-mine ore directly to a continuous and

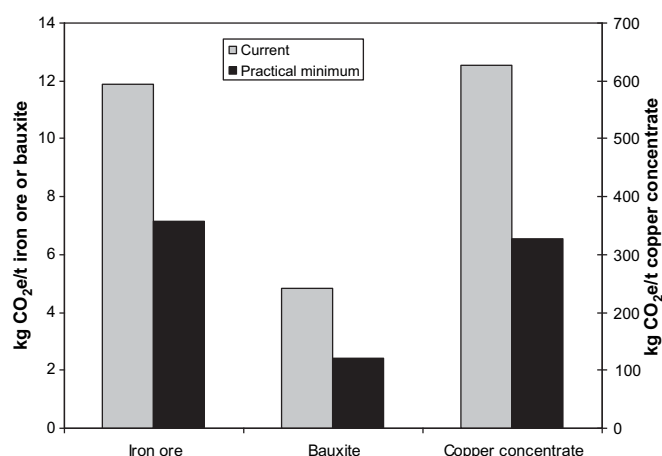


Fig. 8. Comparison of current and best practical minimum GWPs.

dedicated belt conveyor handling system. Potential greenhouse gas savings of 100,000–133,000 t CO₂/y were reported for these systems compared to a conventional shovel/truck operation. Computing applications, both stand-alone and in conjunction with other developing technologies could also contribute to the above energy savings, e.g. in combination with remote sensors to minimise exploration digging and drilling and to measure what is ahead of the mining working face, thereby reducing waste rock handling. Other possible computing applications include optimising the performance of mine ventilation networks [40] and on-line process models to allow plant operators to establish the best operating conditions for the plant.

6. Future demand for metals

Increasing demand for primary metals, together with falling ore grades and more complex ore bodies can be expected to lead to an increase in global energy consumption and greenhouse gas emissions from primary metal production (particularly in the mining and mineral processing stages) in the future, as pointed out earlier. Furthermore, though land used for the extraction of primary metals represents less than 0.1% of the terrestrial surface of the earth [41], exploration and mining activity can affect surrounding ecosystems due to necessary infrastructure and by dispersing metal compounds into the environment, either as air-borne particles or as ions in aqueous solutions. Thus restricting any increased demand for primary metals will help alleviate these associated environmental impacts.

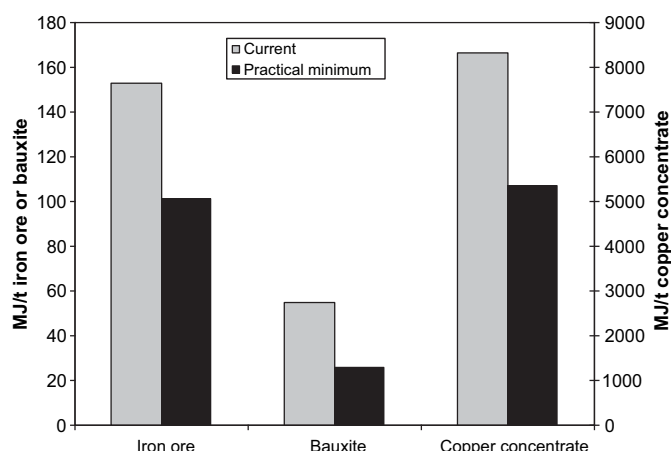


Fig. 9. Comparison of current and best practical minimum embodied energies.

⁴ By applying the % energy savings in Table 4 to the respective energy inputs in Table 2.

While the demand for metals is likely to increase in the future, their usage pattern may well change over the coming decades. As carbon footprints and other environmental impacts of metal production are progressively built into the cost structure of metal production – for example, through carbon taxes – the absolute price of metals will not only increase but there will be a relative shift in price between metals because of the different energy intensities of metal production processes [1]. This will result in greater use of light weight metals in applications where a high strength-to-weight ratio is important, considered over the entire ‘cradle-to-cradle’ life cycle of a metal, for example in land transport applications (cars, trucks) and buildings and other infrastructure. This will see increasing use of aluminium, already a commodity metal, and much greater production and usage of magnesium and titanium which will become commodity metals. However, steel will remain the metal of choice for applications where weight is not an issue as it has the lowest energy intensity (except for lead [1]) and is by far the most abundant metal in nature and occurs in the highest concentration.

One way of reducing the demand for metals and other materials is by dematerialisation. The closure of materials loops through the re-use and recycling of materials complements the process of dematerialisation. Of the materials currently used by society, metals have the greatest potential for unlimited recycling. They are not biodegradable and their elemental nature means that, in principle, they are infinitely recyclable so the stock of ‘metals-in-use’, which is already large will continue to increase. Energy savings of secondary (i.e. recycled) metal production over primary metal production are reported [41–43] to be: copper 85%, nickel 90%, lead 65%, zinc 75%, aluminium 95% and steel 74%. These energy savings will increase in the future for the reasons outlined above. Metal recycling also reduces mining and mineral processing activities that disturb ecosystems.

Thus society has a responsibility to maximise the amount of metals (and other materials) that are recycled if sustainability goals are to be achieved. Two ways being used to help meet this objective are:

6.1. Design for recycling

This is a concept that is increasingly being included in recycling policy and regulations. It is a product design tool that considers the materials from which a product is manufactured and how these materials are assembled. The main criteria to be considered are:

- use recyclable materials – design products using materials that can be recycled;
- use recycled materials – select materials that contain a high percentage of recycled content;
- reduce the number of different materials within an assembly;
- mark parts for simple material identification;
- use compatible materials within an assembly – select materials that do not need to be separated for recycling;
- make it easy to disassemble.

6.2. Extended producer responsibility and stewardship

Extended producer responsibility (EPR) or stewardship programs can be best understood as changing the traditional balance of responsibilities among the manufacturers and distributors of consumer goods, consumers and governments with regard to waste management. Although they take many forms, these programs are all characterized by the continued involvement of producers and/or distributors with commercial goods at the post-consumer stage. EPR extends the traditional environmental responsibilities that producers and distributors have previously been assigned to include management at the post-consumer stage.

The challenge that lies ahead is to devise policies and actions that will continue to help society to optimise the efficient use of metal resources and stocks while at the same time minimising their environmental impacts.

7. Conclusions

Most life cycle assessments of metal production processes do not consider the mining and mineral processing stage in any detail, largely due to a lack of publicly available data and the relatively small contribution that the mining and mineral processing stages make to the ‘cradle-to-gate’ environmental impacts of many metal production processes, particularly with regard to impacts such as embodied energy and greenhouse gas emissions. However, falling ore grades together with the likelihood of more finely-grained and complex deposits in the future will increase the energy and greenhouse impacts of these stages. It is therefore important that the contributions of the various processing steps that make up these stages be quantified, with the major contributing steps being identified in order that efforts to reduce these environmental impacts be focussed on these steps.

Based on the inventory data used in the study, it was observed that loading and hauling made the largest contributions (approximately 50%) to the total greenhouse gas emissions for the mining and processing of iron ore and bauxite (11.9 and 4.9 kg CO₂e/t respectively). In the case of copper ore, it is the crushing and grinding (particularly the latter) steps that make the largest contribution (approximately 46%) to the total greenhouse gas emissions for the mining and processing of copper ore (628 kg CO₂e/t concentrate). These results indicate that efforts to reduce the increased greenhouse gas emissions from mining and mineral processing, anticipated in the future as a result of falling ore grades and more finer-grained deposits, should focus on loading and hauling for iron ore and bauxite, while for copper ore the focus should be on grinding. There are a number of new and emerging technologies that could be expected to assist in this task, and these include high pressure grinding rolls and stirred mills for grinding, and further advances in diesel engine technology for loading and hauling applications, as well as the use of mobile crushing and conveying systems for open-pit mining. Restricting any increase in demand for primary metals in the future through dematerialisation, re-use and recycling will also help alleviate the environmental impacts associated with primary metal production, particularly in the mining and mineral processing stages of the metal life cycles.

References

- [1] Norgate T, Jahanshahi S, Rankin WJ. Assessing the environmental impact of metal production processes. *Journal of Cleaner Production* 2007;15:838–48.
- [2] Norgate T, Jahanshahi S. Energy and greenhouse gas implications of deteriorating quality ore reserves. In: *Proceedings of fifth Australian Conference on Life Cycle Assessment*: Melbourne; 2006.
- [3] Mudd G. An analysis of historic production trends in Australian base metal mining. *Ore Geology Reviews* 2007;32:227–61.
- [4] US Department of Energy. Industrial technologies program. Mining Industry Energy Bandwidth Study; June 2007.
- [5] Fuerstenau D, Abouzeid A-A. The energy efficiency of ball milling in comminution. *International Journal of Mineral Processing* 2002;67:161–85.
- [6] Hilson G. Defining “cleaner production” and “pollution prevention” in the mining context. *Minerals Engineering* 2003;16:305–21.
- [7] Hilson G, Naye N. Environmental management system implementation in the mining industry: a key to achieving cleaner production. *International Journal of Mineral Processing* 2002;64:19–41.
- [8] Macedo AB, de Almeida Mello Freire DJ, Akimoto H. Environmental management in the Brazilian non-metallic small-scale mining sector. *Journal of Cleaner Production* 2003;11(2):197–206.
- [9] Suppen N, Carranza M, Huerta M, Hernandez M. Environmental management and life cycle approaches in the Mexican mining industry. *Journal of Cleaner Production* 2006;14:1101–15.
- [10] Hilson G. Sustainable development policies in Canada’s mining sector: an overview of government and industry efforts. *Environmental Science and Policy* 2000;3(4):201–11.
- [11] Hilson G, Murck B. Sustainable development in the mining industry: clarifying the corporate perspective. *Resources Policy* 2000;26(4):227–38.

- [12] Van Berkel R. Eco-efficiency in primary metals production: context, perspectives and methods. *Resources, Conservation and Recycling* 2007;51:511–40.
- [13] Jenkins H, Yakovleva N. Corporate social responsibility in the mining industry: exploring trends in social and environmental disclosure. *Journal of Cleaner Production* 2006;14:271–84.
- [14] Van Berkel R. Cleaner production in Australia: revolutionary strategy or incremental tool? *Australian Journal of Environmental Management* 2000;7:132–46.
- [15] Mudd G. An analysis of historic production trends in Australian base metal mining. *Ore Geology Reviews* 2007;32:227–61.
- [16] Jolliet O, Muller-Wenk R, Bare J, Brent A, Goedkoop M, Heijungs R, et al. The LCIA mid-point damage framework of the UNEP/SETAC life cycle initiative. *International Journal of Life Cycle Assessment* 2004;9:394–404.
- [17] Environmental management – life cycle assessment – goal and scope definition and inventory analysis. AS/NZS ISO 14041:1999; Standards Australia.
- [18] Yellishetty M, Ranjith P, Tharumarajah A, Bhosale S. Life cycle assessment in the minerals and metals sector: a critical review of selected issues and challenges. *International Journal of Life Cycle Assessment* 2009;14:257–67.
- [19] Durucan S, Korre A, Munoz-Melendez G. Mining life cycle modelling: a cradle-to-gate approach to environmental management in the minerals industry. *Journal of Cleaner Production* 2006;14:1057–70.
- [20] Adachi T, Mogi G. Life cycle inventory for base metal ingots production in Japan including mining and mineral processing processes by cost estimating system database. *Transactions of Nonferrous Metals Society of China* 2007;17:131–5.
- [21] Mangena S, Brent A. Application of a life cycle impact assessment framework to evaluate and compare environmental performances with economic values of supplied coal products. *Journal of Cleaner Production* 2006;14:1071–84.
- [22] Awuah-Offei K, Checkel D, Askari-Nasab. Evaluation of belt conveyor and truck haulage systems in an open pit mine using life cycle assessment. *CCIM Bulletin* August 2009;102:1116 [paper 8].
- [23] Energy efficiency best practice in the Australian aluminium industry. Sector study. Industry, science and resources energy efficiency best practice program; May 2000.
- [24] International Aluminium Institute. Life cycle assessment of aluminium: inventory data for the primary aluminium industry. Year 2005 Update; September 2007.
- [25] Natural Resources Canada. Benchmarking the consumption of Canadian underground bulk mines. <<http://oee.nrcan.gc.ca>>; 2005.
- [26] Australasian Mining and Metallurgy. In: Woodcock J, Hamilton J, editors. 2nd ed. Parkville Australia: The Australasian Institute of Mining and Metallurgy; 1993.
- [27] Mineral Processing Survey. Plants in 2008 survey. *Mining Monthly* 2008 May;91:123.
- [28] International Iron and Steel Institute. World steel life cycle inventory, www.worldsteel.org; 2000.
- [29] Labriola A. Unearthing the carbon footprint. *Australian Mining* 2009;101(3):34–5.
- [30] International Iron and Steel Institute. Steel statistical yearbook, www.worldsteel.org; 2008.
- [31] International Copper Study Group. ICSG Press Release: New Edition of 'Directory of Copper Mines and Plants', www.icsg.org; 2009.
- [32] Norgate TE, Jahanshahi S, Rankin WJ. Alternative routes to stainless steel – a life cycle approach. In: Proceedings of Tenth International Ferro Alloys Congress (INFACON X). Cape Town, South Africa; 2004. pp. 693–704.
- [33] Norgate TE, Rankin WJ. Greenhouse gas emissions from aluminium production – a life cycle approach. In: Proceedings of Greenhouse Gases in the Metallurgical Industries: policies, abatement and treatment, COM2001. Toronto; 2001. pp. 275–90.
- [34] Norgate TE, Rankin WJ. Life cycle assessment of copper and nickel production. In: Proceedings of MINPREX 2000. Melbourne; 2000. pp. 133–8.
- [35] Tavares L. Particle weakening in high-pressure rolls grinding. *Minerals Engineering* 2005;18:651–7.
- [36] Sinnott M, Cleary P, Morrison R. Analysis of stirred mill performance using DEM simulation: part 1 – media motion, energy consumption and collisional environment. *Minerals Engineering* 2006;19:1537–50.
- [37] Brent, G. Reducing energy and greenhouse gas intensities of mining through blasting. Sustainable development conference, Darwin. Minerals Council of Australia, www.sd08.com.au; September 2008 [accessed 17.03.09].
- [38] Anon. Mobile and continuous. *Mining Magazine* May 2009:46–8.
- [39] Lucio J, Senra C, Souza A. Paving the future – a case study replacing truck-and-shovels by shovel-and-conveyor continuous mining at Carajas open pit mines. In: Proceedings of iron ore conference 2009: Perth; 2009. pp. 269–76.
- [40] Lowndes I, Fogarty Z, Yang Z. The application of genetic algorithms to optimise the performance of a mine ventilation network: the influence of coding method and population size. *Soft Computing* 2005;9:493–506.
- [41] Wernick I, Themelis N. Recycling metals for the environment. *Annual Reviews of Energy and the Environment* 1988;23:465–97.
- [42] Norgate TE, Rankin WJ. The role of metals in sustainable development. In: Proceedings of green processing 2002. Cairns; 2002. pp. 49–55.
- [43] Hagelstein K. To air is human. In: Proceedings of John Floyd international symposium on sustainable developments in metals processing. Melbourne, 2005. pp. 49–60.